Generalization preserves tableaux derivability for fresh constants

The Open Logic Project contributors

openlogic:first-order-logic/tableaux:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof embeds two before/after tableau diagrams through the raw tableau environment (lines 1243-1255); smoke-tested to compile and render as trees with the brace argument consumed silently

Editorial note. The label thm:strong-generalization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Eigenvariable condition for the False-forall rule. From False (forall x A(x)) the False-forall rule derives False A(c) for a constant c that does not occur above the inserted line in the tableau.. The inserted line False A(c) is a correct rule application exactly because c does not occur in the assumptions B_1,...,B_n nor in forall x A(x).
- Derivability via closed tableaux. Gamma proves A iff for some finite B_1,...,B_n in Gamma the set {False A, True B_1,...,True B_n} has a closed tableau.. The hypothesis unpacks into a closed witness tableau whose first assumption is replaced in the construction.

Theorem 0.1. *If c is a constant not occurring in Γ or $A(x)$ and $\Gamma \vdash A(c)$, then $\Gamma \vdash \forall x\, A(x)$.*

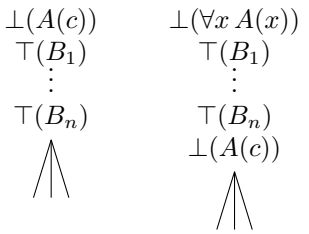
Proof. Suppose $\Gamma \vdash A(c)$, i.e., there are B_1, ..., $B_n \in \Gamma$ and a closed tableau for

$$\{\bot(A(c)), \top(B_1), \ldots, \top(B_n)\}.$$

We have to show that there is also a closed tableau for

$$\{\bot(\forall x\, A(x)), \top(B_1), \ldots, \top(B_n)\}.$$

Take the closed tableau and replace the first assumption with $\bot(\forall x\, A(x))$, and insert $\bot(A(c))$ after the assumptions.



The tableau is still closed, since all sentences available as assumptions before are still available at the top of the tableau. The inserted line is the result of a correct application of $\bot\forall$, since the constant c does not occur in B_1, …, B_n or $\forall x\, A(x)$, i.e., it does not occur above the inserted line in the new tableau. □